L2.4 Limits at Infinity

§3.4 — where the asymptote rules come from

The three asymptote rules

You already know the left column — it is precalc. The right column is today.

The rule	Where it comes from
① **BOT > TOP** HA $y = 0$ e.g. $\frac{x^2+x}{x^3}$	
② **BOT = TOP** HA at the ratio of leading coefficients e.g. $\frac{3x^2-1}{4x^2}$	
③ **TOP > BOT** no HA; SA by polynomial long division e.g. $\frac{4x^2-4}{3x-6}$	

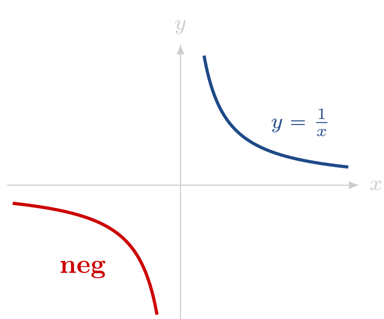
THE LIMIT AT INFINITY THEOREM — all three rules run on this one

$$\boxed{\lim_{x\to\infty} \frac{a}{x^r} = 0 \qquad a \text{ real}, \quad r \text{ a positive rational}}$$

Also true as $x \to -\infty$, whenever x^r is defined there.

"divide top and bottom by the highest power of x in the BOTTOM, then kill every term that turned into a over x to a power"

$\sqrt{x^2} = |x|$ — the sign on the way to $-\infty$



A square root is never negative, so $\sqrt{x^2}$ is not x — it is $|x|$.

$$\boxed{x < 0 \implies \frac{1}{x} = -\frac{1}{\sqrt{x^2}}}$$

"on the way to $-\infty$ the $\frac{1}{x}$ is negative, and the radical is not, so a minus sign has to appear when you move it inside"

SLANT ASYMPTOTE — and why the line is there

Long division on $\dfrac{4x^2-4}{3x-6}$ gives

$$\boxed{\frac{4x^2-4}{3x-6} = \underbrace{\tfrac{4}{3}x+\tfrac{8}{3}}_{\to +\infty} + \underbrace{\frac{12}{3x-6}}_{\to 0} \qquad SA = \tfrac{4}{3}x+\tfrac{8}{3}}$$

"the remainder dying is the whole reason the line is an asymptote — the limit is $+\infty$ and the answer is still a line"

$\infty - \infty$ — rewrite first

∞ is not a number, so the limit laws do not apply to it. Rewrite, *then* take the limit. For a difference of two roots, or a root minus a polynomial, multiply by the conjugate over itself — you have done conjugates before.

$$\boxed{\lim_{x\to\infty}\left(\sqrt{x^2+1}-x\right) = \lim_{x\to\infty}\frac{1}{\sqrt{x^2+1}+x} = 0}$$

NOTATION

HA horizontal asymptote **VA** vertical asymptote **SA** slant asymptote
Z@ zero at **@** at **TOP** / **BOT** the degrees, top and bottom

WATCH OUT

✗ ~~$\infty - \infty = 0$~~ → **∞ is not a number — rewrite, then take the limit**
✗ ~~divide by the highest power on top~~ → **by the highest power in the BOTTOM**
✗ ~~$\sqrt{x^2} = x$~~ → **$\sqrt{x^2} = |x|$, and on the left that is $-x$**
✗ ~~a graph has one horizontal asymptote~~ → **it can have two — one per side, as $\frac{\sqrt{2x^2+1}}{3x-5}$ does**
✗ ~~the graph can't cross a horizontal asymptote~~ → **it can. An HA is end behaviour, not a barrier**
✗ ~~no HA means no asymptote~~ → **TOP one degree higher $\Rightarrow$ SA, by long division**
✗ ~~TOP > BOT always gives a slant asymptote~~ → **only when the degrees differ by exactly one**
✗ ~~l'Hôpital gives me the asymptote~~ → **it gives the limit. The division gives the ratio of leading coefficients, which is the line**